# Arnoneel Sinha, PhD

*Post-Doctoral Fellow, CBGM, Augusta University*

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## Professional Summary

Biostatistician and postdoctoral fellow specializing in single-cell and multi-omics analysis, statistical methodology, and reproducible computational workflows, with active work on human-audited, agentic AI systems for biomedical discovery. Over five years of experience building end-to-end scRNA-seq and bulk RNA-seq pipelines in R and Python, co-authoring peer-reviewed manuscripts and NIH proposals, and teaching graduate biostatistics and computational genomics. Current projects include cellivista Nexus, a local-first platform that pairs deterministic Seurat-based analysis with retrieval-augmented, approval-gated AI agents and full provenance tracking. Seeking to lead an independent research program in statistically rigorous, trustworthy AI-assisted methods for biomedical data.

## Skills

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| **Technical Skills** | **Statistical Methods**: Regression and GLMs, ANOVA, experimental design, survival/time-to-event analysis, mixed-effects and time-series models, Bayesian inference, changepoint detection.<br>**Genomics & Multi-omics**: Bulk and single-cell RNA-seq (QC, normalization, clustering, differential expression, cell-type annotation, cell-cell communication, metabolic flux), proteomics (mass spectrometry), cross-platform data integration. Tools: Seurat, DESeq2, Cell Ranger, STAR, HTSeq, Bioconductor, GSEA, Enrichr.<br>**Machine Learning**: Supervised and unsupervised methods applied to biomedical data.<br>**Data Science**: Large-scale data cleaning and integration; visualization with ggplot2, Shiny, and matplotlib; statistical analysis plans for collaborative studies and grants. |
| **Programming & Languages** | R (Seurat, DESeq2, ggplot2, Shiny), Python (pandas, scikit-learn, matplotlib), SAS, LaTeX |
| **Computing Environments** | High-performance computing (SLURM), Linux/Bash, version control (Git) |
| **Languages** | English (fluent), Hindi (fluent), Bengali (native) |

## Professional Experience

**Post-Doctoral Fellow** **Aug 2023 - Present**
*Dr. Richard McIndoe lab, Center for Biotechnology & Genomic Medicine, Augusta University* *Augusta, GA*

- Analyzed 15+ single-cell and bulk RNA-seq datasets across tumor, pancreatic, and other tissue studies, building reusable R pipelines for differential expression, cell-type annotation, and pathway enrichment (Seurat, DESeq2, Loupe Browser).
- Provided statistical and bioinformatics support to interdisciplinary teams (R, Python); co-authored peer-reviewed manuscripts and contributed analysis plans to NIH and NSF grant proposals.
- Mentored graduate students on computational genomics projects, from study design through analysis and interpretation.

**Graduate Research Assistant** **Aug 2018 - July 2023**
*Augusta University* *Augusta, GA*

- Delivered statistical consulting across multidisciplinary studies and taught undergraduate mathematics and statistics.
- Applied machine learning and advanced modeling to clinical trial data.

**Senior Subject Matter Expert** **Dec 2017 - Mar 2018**
*Nerdy Turtlez* *Kolkata, India*

- Ran client data-analytics and visualization projects, performing statistical analyses and building the accompanying reports and presentations.

**Research Mentor, Undergraduate Training Program** **2022-2023**
*Statistics Undergraduate Research Experience (SURE), Southern Regional Council on Statistics (SRCOS)*

- Mentored undergraduate STEM majors during the SURE Summer Research Conferences (SRCOS 2022 and 2023), providing guidance on graduate school and career opportunities in statistics.
- Collaborated with session leaders to facilitate discussions and address student questions during mentoring sessions.

## Teaching Experience

Taught graduate and undergraduate courses in statistics, mathematics, and computational genomics (in-person and hybrid), covering lecture delivery, grading, and one-on-one mentoring.
**Graduate Courses:**

**Co-Instructor, Translational Genomics and Proteomics (GNMD 8051)** Spring 2025, 2026

- Co-instructed a graduate-level course for 2nd year biomedical students on advanced genomic and proteomic technologies and their applications in translational medicine
- Delivered a lecture on genomic approaches to disease classification and prediction
- Supported students in critical reading of omics literature, project-based assignments, and discussions on emerging genomic tools

**Co-Instructor, Computational Methods in Genomics and Genetics (GNMD 8050)** Fall 2024, 2025, 2026

- Co-taught a 4-credit graduate-level course combining lectures and supervised labs, introducing students to computational and statistical techniques in genomic and genetic research
- Led multiple sessions covering R programming, data visualization, statistical methods, and machine learning techniques including regression, clustering, dimensionality reduction, and neural networks
- Collaborated with faculty to design and evaluate student projects and assessments, and supported hands-on learning through case-based data analysis

**Undergraduate Courses:**

- **Elementary Statistics (MATH 1401)**
  Hypothesis testing, regression, ANOVA, CLT, etc. Fall 2021
- **Pre-Calculus Mathematics (MATH 1113)**
  Polynomial/exponential functions, trigonometry, unit circle Fall 2020, Spring 2021

## Publications & Presentations

### Publications

### Peer-Reviewed Publications

- Kimberly A. Rivera-Caraballo, Tae Jin Lee, **Arnoneel Sinha**, Marco Orecchioni, Rafal Pacholczyk, Karina Vázquez-Arreguín, Shilpa Sharma, Kimya Jones, Kailash Vemuri, Upasana Sahu, Sara A. Murphy, Bangxing Hong, Ravindra Kolhe, Ashok Sharma, Balveen Kaur; Oncolytic HSV-1-Mediated JAG1 Blockade Induces Glioma Senescence-Associated Secretory Phenotype to Increase Macrophage Activation and Cetuximab-Mediated Senolysis. Cancer Res 2025; https://doi.org/10.1158/0008-5472.CAN-25-1402
- James JN, Frazier KB, Looney SW, **Sinha A**, Faigen AB. *Does a Difference Exists in Comprehensive Basic Science Examination Scores of 4-Year versus 6-Year Oral-Maxillofacial Surgery Residents?* J Oral Maxillofac Surg. 2020 Sep;78(9):1459.e1-1459.e6. doi: 10.1016/j.joms.2020.04.007. Epub 2020 Apr 18. PMID: 32413336.

**In Preparation**

- **Sinha A**, Sinha K., McIndoe R. *cellivista: An Interactive Seurat-Based R Shiny App for Single-Cell RNA-seq Analysis.* (Submitted to The Journal of Open Source Software)
- **Sinha A**, Chen J. *Bayesian Multiple Change Point Detection for Bivariate Negative Binomial Sequences with Application to Integrative Analysis of Paired RNA and ChIP Sequencing Counts for ENCODE K562 Cell Line.* (Submitted to *Journal of Computational Biology*)
- **Sinha A**, Chen J. *Bayesian Change-Point Detection in Bivariate Pseudo-Time Trajectories using Single-Cell RNA Sequencing Data.*
- **Sinha A**, Sinha K, Lu S, Chatterjee K. *A Machine Learning Enhanced Exponentially Weighted Moving Average Maximum Chart for Joint Monitoring of Risk-Adjusted Continuous Outcomes* (Submitted to *Quality and Reliability Engineering International*)
- **Sinha A**, Sinha K, Chatterjee K. *Comparative Sequential Monitoring of Mixed and Serially Correlated Multivariate Processes After a Common Transformation Framework*

**Presentations**

- **Arnoneel Sinha**, Richard A. McIndoe. *cellivista: An Interactive Seurat-Based R Shiny App for Single-Cell RNA-seq Analysis.* **Graduate Research Day**, Augusta University, GA, March 2026. (Poster presentation)
- **Arnoneel Sinha**, *Biostats for Flow Cytometry Users: Designing With Statistical Sense*, **Southeast Flow Cytometry Interest Group**, Atlanta, March 2026.
- **Arnoneel Sinha**, *Opportunities for Statisticians & Biostatisticians: From India to a Global Career*, **One Day Talk (Online Mode)**, Trivenidevi Bhalotia College, India, November 2025. (Virtual talk)
- **Arnoneel Sinha**, Richard A. McIndoe. *cellivista: An Interactive Seurat-Based R Shiny App for Single-Cell RNA-seq Analysis.* **Health Sciences Fair and Expo**, Augusta University, GA, 2025. (Poster presentation)
- **Arnoneel Sinha**, Jie Chen. *Detecting Multiple Change Points in Bivariate Negative Binomial Model with Application to COVID-19 Incidence and Death Counts.* **Summer Research Conference, Southern Regional Council on Statistics**, Waco, TX, 2023. (Poster presentation)
- **Arnoneel Sinha**, Jie Chen. *Detecting Multiple Change Points in Bivariate Negative Binomial Model with Application to COVID-19 Incidence and Death Counts.* **The 3rd Annual Workshop: Emerging Data Science Methods for Complex Biomedical and Cyber Data**, Augusta University, Augusta, GA 2023. (5 minute presentation)
- **Arnoneel Sinha**, Jie Chen. *Detection of Multiple Change Points in Bivariate Data: Application to COVID-19 Incidence and Death Rates.* **Summer Research Conference, Southern Regional Council on Statistics**, Jekyll Island, GA, 2022. (Poster presentation)

# Education

**Doctor of Philosophy (Biostatistics)** **Aug 2018 - July 2023**
*Augusta University*

**Dissertation Title:** On the Detection of Multiple Change Points in a Sequence of Bivariate Random Vectors with Application to COVID-19 Data

**Master of Science (Statistics)** **2015 - 2017**
*Visva Bharati University*

**Thesis Title:** An overview of Process Capability Analysis

**Bachelor of Science (Statistics)** **2012 - 2015**
*Visva Bharati University*

**Thesis Title:** Juvenile Apprehension in India (2001-2012): A Statistical Perspective

## Academic Projects & Theses

**Selected Academic Projects & Theses**

- **A Review on Gene Set Analysis**

  *'Omics Data Analysis' course final project*

- **A Comparison of Five Machine Learning Techniques Predicting Time to Failure in Laboratory Earthquakes**

## Honors, Awards & Professional Service

**Awards & Honors**

- 'Award of Excellence in Research' (1st place for poster presentation), Graduate Research Day, Augusta University (March, 2026)
- Winner of '*People's Choice Award*' in the 3MT competition at Augusta University (March, 2023)
- '*NSF Travel Award*' winner at the 3rd Annual Data Science Workshop, Augusta University (March, 2023)
- '*Boyd Harshbarger Travel Award*' winner at the Summer Research Conference, SRCOS (April, 2022)
- '*Outstanding Student of the Year*' award at Biostatistics & Data Science Division, Augusta University (2021-2022)

**Professional Service & Membership**

- Member of the *Augusta University Post-Doctoral Association*, Augusta University (Fall 2024 - Present)
- Member of the *Student Advisory Council*, Enrollment & Student Affairs, Augusta University (Fall 2024 - Present)
- Member of the *Graduate Student Council* (GSC), Augusta University (Fall 2021 - Summer 2023)
- GSC representative on the *The Graduate School - Faculty Council*, Augusta University (Fall 2022 - Summer 2023)

## Training / Certificates

**One year Master of Business Administration (MBA)** **2017**
*National Institute of Business Management (NIBM)*

Specialization: Market Research

**Business Analytics training course** **2017**
*CONCEPTlytics Institute & Technologies*